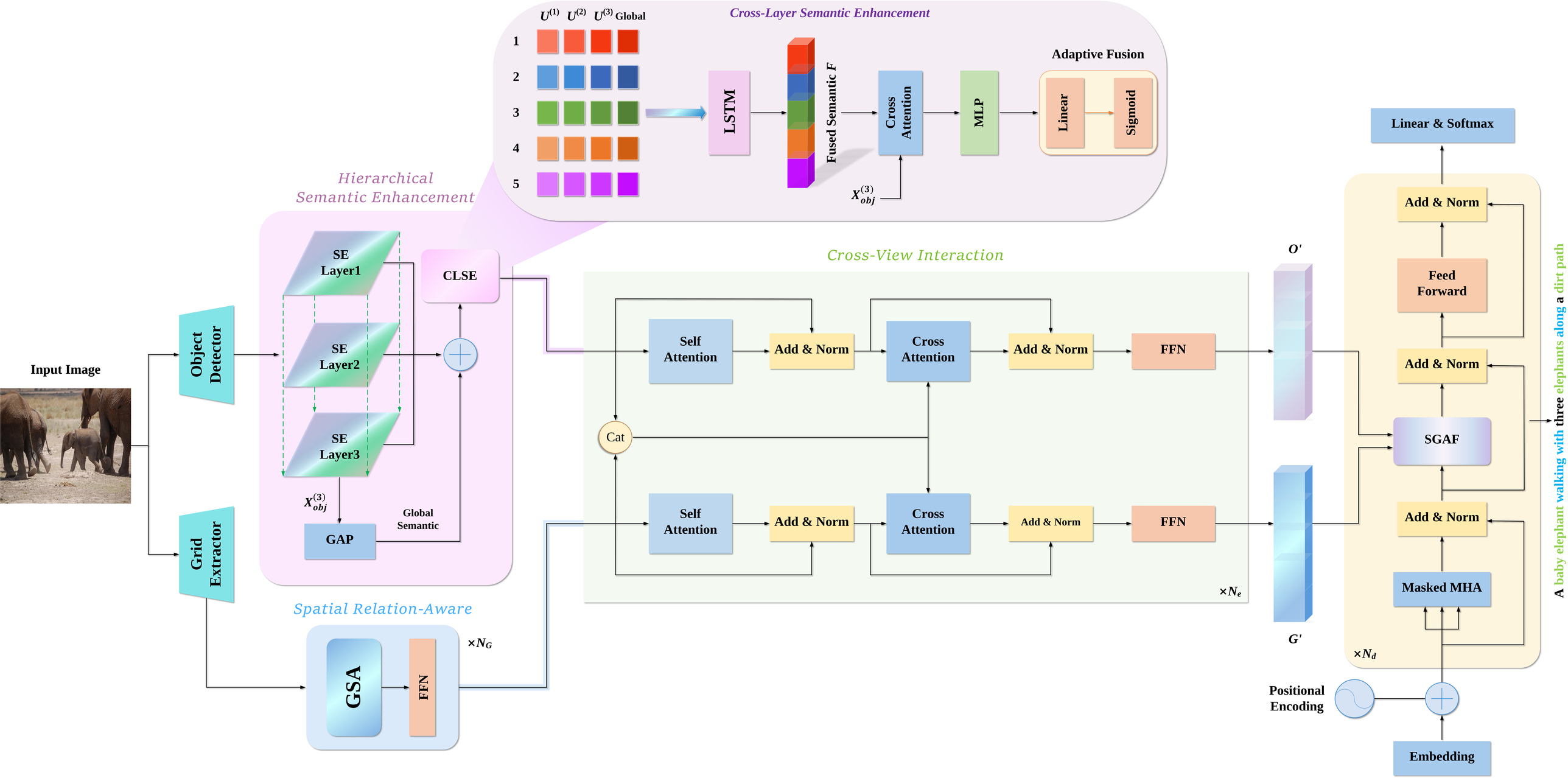

Input Image
Object Detector
Grid Extractor
Hierarchical Semantic Enhancement
SE Layer1
SE Layer2
SE Layer3
$X_{obj}^{(3)}$
GAP
Global Semantic
CLSE
Cross-Layer Semantic Enhancement
$U^{(1)}$ $U^{(2)}$ $U^{(3)}$ Global
1
2
3
4
5
LSTM
Fused Semantic F
Cross Attention
MLP
Adaptive Fusion
Linear
Sigmoid
$X_{obj}^{(3)}$
Cross-View Interaction
Self Attention
Add & Norm
Cross Attention
Add & Norm
FFN
Cat
Self Attention
Add & Norm
Cross Attention
Add & Norm
FFN
$\times N_e$
Spatial Relation-Aware
GSA
FFN
$\times N_G$
O'
G'
Linear & Softmax
Add & Norm
Feed Forward
Add & Norm
SGAF
Add & Norm
Masked MHA
$\times N_d$
Positional Encoding
Embedding
A baby elephant walking with three elephants along a dirt path